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SHORT-PAPER

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Pervasive Games in the Rehabilitation Process: A Systematic Literature Review

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ABSTRACT

Videogames have gained great interest in the computer science community, this interest has expanded to the area of health. This situation is because the results obtained with gaming experiences in the rehabilitation therapies. Most rehabilitation activities (such as physical and mental exercises) are monotonous and boring for patients. It is essential to generate and maintain the motivation of patients. The Pervasive Games (PG) is a proposal that helps patients to increase and maintain their motivation. This article presents an exhaustive review of 101 final studies, which seeks to identify the specific use of PG in the rehabilitation of patients. The objective of the review is to identify and classify the applications that are given to the PGs within the area of health. This research will help to propose a basis and strengthen the use of PG as a rehabilitation tool in some procedures.

CCS CONCEPTS

• **Software and its engineering → Interactive games**.

KEYWORDS

Pervasive games, Rehabilitation, Patient Motivation, Patient Engagement, Literature Review

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1 INTRODUCTION

The rehabilitation of patients with physical and mental problems is an important process in the immediate future for the quality of life of these people. The purpose of rehabilitation is to improve the movement of patients, so, they can independently perform daily tasks on their own [8]. In the field of health sciences, rehabilitation is considered a transcendent point since it depends on how a person will recover after a surgical procedure or how it will evolve in the middle of a therapeutic treatment. This is a complex process due to the psychological affectation that people present in this situation. Even more difficult than generating motivation is to maintain it throughout the time of therapies, treatments or rehabilitation activities. This is why it is considered that a tangible-virtual environment for rehabilitation is feasible and that this approach is highly appreciated by the patients [9].

On the other hand, videogames have gained great interest in the computer science community, to such that this interest has been extended to other areas of knowledge such as health. This is the positive consequence of the use of these gaming experiences as support for rehabilitation therapies for people with physical and mental difficulties. The majority of traditional rehabilitation activities (such as physical and mental exercises) are monotonous and boring for patients [3], which does not allow these people to advance in recovery. For this reason, it is essential to generate and maintain the motivation of patients in rehabilitation processes.

Videogames, and even more PG provide a better user experience, which generates greater motivation in the interaction. In addition, this kind of games allows in different ways the incorporation and integration of the real world and the virtual world.

For this reason, we carried out a systematic literature review (SLR) to understand the way in which games are used in rehabilitation. In this specific case we evaluated the use of PG as a support tool in rehabilitation processes. In addition, it is planned to identify the rehabilitation processes that most use PG and the associated technologies with this type of games. Previous studies have show successful results to the scientific community in a general field [1] and educational context as well [2].

This article is structured in the following way. In section 2, we describe the main concepts that are used in rehabilitation through

games. In section 3, we present the methodology that was used to carry out this study. In section 4, the results of the study are discussed. Finally, the conclusions and future work are established in section 5.

2 BACKGROUND

The concepts of rehabilitation, game and PGs are presented below. In addition, we indicate the analysis carried out and the discussion about the research problem. Finally, the different works related to the classification and application of PGs in-patient rehabilitation environments are shown.

2.1 Rehabilitation

Rehabilitation assists patient to achieve functional Independence, and improve their motor skills [6]. The rehabilitation is known with the set of techniques and methods that serve to recover a function or activity of the body that has been reduced or lost due to an accident or illness. This process is performed on patients to improve their quality of life, whether in mobility and / or mentality. The traditional way in which the recovery activities of a patient are carried out becomes boring and annoying for many of them, which generates a decrease in the rehabilitation activity.

2.2 Games

Games have been present in the lives of people for a long time. Traditional games involved the use of physical elements by players. Time passed and videogames generated a revolution in this field and attracted a good number of new players. Some of the keys to the success of videogames with respect to traditional games are: the creation of the illusion of being immersed in a virtual world with graphics and sound by computer, the definition of objectives typically more interactive than those of traditional games and the design of games with an optimal level of complexity, which can provoke the curiosity of the players in an easy way.

2.3 Pervasive Games

Pervasive games represent a radically new game form that transfers gaming experiences out into the physical world, weaving ICTs into the fabric of players' real environments [5]. Many different forms of gaming have been grouped under the concept, including the massively collaborative troubleshooting games (The A.I. Game), the location-based mobile games (Botfighters), the games augmenting the reality with ludic content (Visby Under) and the games staged with a combination of virtual and physical elements [7].

Pervasive games represent a rapidly growing field of research, as evidenced from the proliferation of available prototypes [4]. In this study, we want to delve into the field of PGs used in rehabilitation processes to identify their impact, their characteristics and the main areas used by doctors and physiotherapists.

3 SYSTEMATIC LITERATURE REVIEW

The purpose of this work was to identify the use that PGs are given within the field of health, specifically in the physical and mental rehabilitation of patients. We carried out a categorization of the technologies used in the PGs for each rehabilitation process. Finally, taking into account the different expansions of the pervasiveness, we present the health areas that use each one of these with greater frequency. In this way, the research questions are formulated below.

3.1 Research Questions

- RQ1: Are the PGs used in the patient rehabilitation processes?
- RQ2: If the PGs are used, What are the most used technologies for this purpose and how are they implemented?
- RQ3: What are the areas of health that commonly use PGs as a support tool and how they relate to each level of pervasiveness?

3.2 Digital Databases

In order to search for relevant information in the research, it was necessary to evaluate a set of digital databases. The topics considered to filter the databases were the areas of videogames, technology, computer science, therapies, health and rehabilitation. The chosen databases are shown in Table 1.

Table 1: Selected digital databases

ID	Name	URL
DB1	Scopus	https://www.scopus.com/
DB2	ACM Digital Library	https://dl.acm.org/
DB3	DOAJ	https://doaj.org/
DB4	Springer Link	https://link.springer.com/
DB5	IEEE Xplore	http://ieeexplore.ieee.org/
DB6	Web of Science	https://webofknowledge.com/

3.3 Query Strings and Criterions

In order to search the information in the databases defined in the Table 1, it was necessary to incorporate the query strings to executed in each database. The strategy for the construction of the query strings is based on the definition of the main terms and their synonyms. The search has been limited in articles published since 2013 to the present, in order to better analyze the most recent results. The base of the query strings that were built are exposed below.

((game OR pervasive) AND ((healthcare OR therapy) AND rehabilitation)

In addition, a set of inclusion and exclusion criterions were defined to filter the results.

- Inclusion Criterions
  - Paper published between 2013 and current date.
  - Paper published as result of conferences, congress or journals.
  - Paper written in English.
  - Paper included into databases shown on Table 1.
  - Paper associated with these topics: Health, Pervasive Games, Rehabilitation and related.
- Exclusion Criterions
  - Paper only with content table or resume.

- Paper not related to a research.
- Paper not related to Health, Pervasive Games, Rehabilitation and related.
- Short papers.
- Paper which does not fulfill at least one of inclusion criterions.

3.4 Data Extraction

The query strings execution yielded a total of 1849 records. Which are distributed in the different databases as shown in Table 2.

Table 2: Results of query strings execution.

Database ID	Amount of Papers
DB1	769
DB2	82
DB3	12
DB4	326
DB5	351
DB6	309
Total	**1849**

4 DATA ANALYSIS AND RESULTS

First, having the complete list of results, we analyze each of them to identify indications to filter the articles. The length of these was verified to separate short articles, tables of contents, duplicate articles and indexes. In this process, 615 articles were discarded (Figure 1).

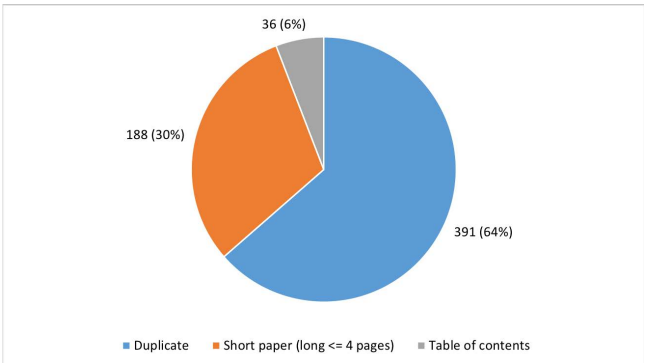


Figure 1: Results about first set of filters applied in results.

In Figure 1 it is evident that the same paper is indexed in different databases, which generates duplicity, these duplicate items represent 64% (391) of the articles rejected in the first filter. Secondly, we rejected 188 (30%) short papers. Finally, 6% (36) of the papers in this first filter were rejected as tables of contents or indexes.

4.1 Applying Additional Filters

We analyzed each paper found evaluating its title and abstract. We read these sections, checking that each one had relation with the topic research. The amount of accepted papers for each filter is presented in Table 3.

Table 3: Results later title, abstract and fulltext filters were applied.

Database ID	Initial Amount	Title	Abstract	Fulltext
DB1	39	138	44	24
DB2	407	226	48	6
DB3	9	167	69	1
DB4	312	23	19	11
DB5	263	8	5	39
DB6	204	54	21	20
Total	**1234**	**616**	**206**	**101**

As table 3 presents, when the filters of title, abstract and fulltext were applied, these generated 616, 206 and 101 articles respectively. From this last amount, a deep analysis was generated to solve the research questions. In this phase, we were sure of the relevance of each article.

4.2 Research Questions Answers

Based on the SLR and the analysis of results, the answers to the research questions in this study are presented below.

4.2.1 RQ1. In the SLR different studies were found, which show the work that has been done in the rehabilitation processes with the support of the PG. However, in many studies, these studies are not defined as PGs, wherefore it was necessary evaluate the pervasiveness properties to identify them. As can be seen in Figure 2, articles related to PGs in different areas of rehabilitation were found.

It is possible to distinguish the importance of PG in rehabilitation processes of stroke patients with 25% of the findings. Also, patients who are in general physical therapy are an important part of this study with 29%. Followed by rehabilitation in the arms (16%) and patients with cerebral palsy (11%). In this way it can be stated that PGs are used in the rehabilitation processes of patients with physical and mental problems.

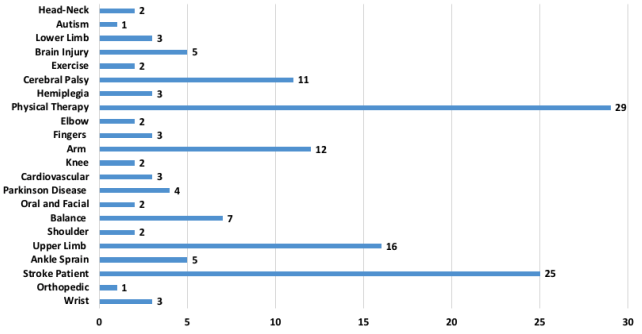


Figure 2: Amount of papers for each subcategory of the rehabilitation category.

4.2.2 RQ2. As mentioned above, PGs are widely used in the rehabilitation processes of patients with physical and mental disabilities. Also, this study allowed us to classify the types of games that are most relevant in these rehabilitation processes (Figure 3). In that figure, it is possible to identify types of PG and implementation technologies used to improve the user experience throughout rehabilitation process.

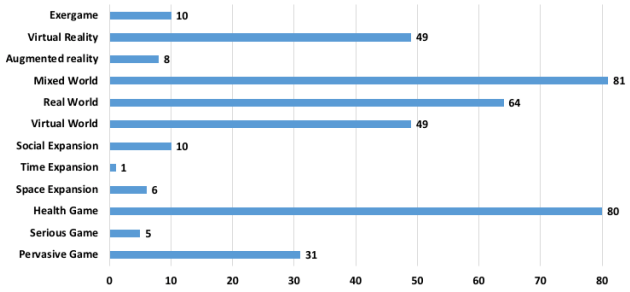


Figure 3: Amount of papers for each subcategory of the game category.

In this same line, several studies are related to the increase of motivation in therapy sessions and procedures related to rehabilitation. This is why the close relationship between these two areas of knowledge is demonstrated in order to improve the quality of life of patients and accelerate rehabilitation compared to traditional techniques.

On the other hand, according to the analysis carried out, the implementation of the PGs in the rehabilitation processes is done through the experience of the doctors and/or physiotherapists. There is no a general model that allows defining the type of PG or pervasiveness recommended in a specific case of recovery. However, the success of this union of knowledge areas is undeniable.

During the classification process, we could also identify an effort that is being made so that the rehabilitation activities are done from home. In this way. It is demonstrated that although therapies were previously carried out in specialized centers, nowadays the comfort of the home is being sought to perform these activities.

4.2.3 RQ3. As aforementioned in the RQ1 response, PGs are widely used in the rehabilitation processes of patients with physical and mental disabilities. However, there are differences between the amounts of studies carried out between one and another area, It is possible to affirm that the largest number of studies (identified throughout the investigation) focus on problems related to the rehabilitation of patient stroke, cerebral palsy patients and rehabilitation of upper limbs. These special cases are related to spatial and social pervasiveness, leaving aside the temporal pervasiveness (only 1 study considered it). Taking a look at the technological devices, the sensors are a fundamental part of these for the location and identification of the movements of the patients.

5 CONCLUSIONS AND FUTURE WORK

In this article, we have performed an SLR on the processes, devices and application of PGs in the rehabilitation activities of patients with physical and mental disabilities, which have strongly attracted the interest of researchers in the fields of computing and health. Advances and collaborative work have given successful results in terms of improving patients. In this way, the evolution of technology and PGs have allowed people from the health area to apply them as an interaction tool to increase motivation in rehabilitation activities.

In this line, it can be concluded that PGs are widely used in the rehabilitation processes of patients, in addition, the specificity that occurs in the use of technological devices is important, each of them associated with the processes that are carried out in different body and mental parts of the patient. Also, forms of pervasiveness have been identified are used in the majority of the studies found.

With the progress in the rehabilitation processes, there is no doubt that it is a field of research where pervasiveness should be deepened and applied to a greater degree, which would provide a higher level of motivation to patients, positively impacting their results of recovery.

As future work, it is proposed to carry out an investigation with experts in the area of rehabilitation with the objective of identifying and classifying the optimal PG options for each case of therapy. In this study we must consider the elements previously identified, and the experience of these people. In addition, it is important to delve into the impact that is had in the use of these technologies in each of the age ranges, because each individual can react differently to the same technology.

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